Module 3 –

A. Data Link Layer & Framing

1. What is the Data Link Layer, and what are its main functions?

The **Data Link Layer** is the second layer of the OSI model. It provides **node-to-node (hop-to-hop) delivery** of data.

Main functions:

- Framing
- Flow control
- Error control
- Physical addressing
- Medium access control

2. What is the difference between DLC and MAC?

- **DLC (Data Link Control):** Handles framing, flow control and error control.
- **MAC (Media Access Control):** Controls access to a shared transmission medium and handles MAC addressing.

3. What is framing, and why is it required?

Framing is the process of dividing a stream of bits into manageable units called **frames**.

It is required so that the receiver can identify the **beginning and end of each data unit**.

4. What is the difference between fixed-size and variable-size framing?

- **Fixed-size:** Every frame has the same size.
- **Variable-size:** Frames can have different sizes and need additional information to identify their boundaries.

5. What is byte stuffing?

Byte stuffing is used in character-oriented protocols.

If a special flag byte appears inside the data, an **escape (ESC) byte** is inserted before it so that it is not mistaken for a frame boundary.

6. What is bit stuffing?

Bit stuffing is used in bit-oriented protocols.

Whenever the sender finds **five consecutive 1s** in the data, it automatically inserts a **0** after them.

The receiver removes this extra 0.

7. What is flow control?

Flow control controls the rate at which a sender transmits data so that a receiver is not overwhelmed.

It prevents **receiver buffer overflow**.

8. What is the difference between point-to-point and broadcast links?

- **Point-to-point:** Connects two devices directly.
- **Broadcast:** A single transmission medium is shared by multiple devices.

B. Error Detection & Correction

9. What is an error? Explain single-bit and burst errors.

An **error** occurs when received data differs from transmitted data.

- **Single-bit error:** Only one bit is changed.
- **Burst error:** Two or more bits within a data unit are affected.

10. What is the difference between error detection and correction?

- **Error detection:** Determines whether an error has occurred.
- **Error correction:** Determines the error and corrects the affected data.

11. What is redundancy?

Redundancy means adding extra bits to the original data.

These additional bits help the receiver **detect or correct errors**.

12. What is Hamming distance?

Hamming distance is the number of bit positions in which two codewords differ.

Example:

```
10101
10011
```

They differ in 2 positions, so:

Hamming distance = 2

13. What is minimum Hamming distance?

The **minimum Hamming distance (dmind_{min})** is the smallest Hamming distance between any two valid codewords.

It determines the error-detection and error-correction capability of a code.

14. What is a parity-check code?

A **parity bit** is added to data to make the number of 1s either even or odd.

It can detect many **single-bit errors**, but it cannot reliably detect all multiple-bit errors.

15. What is CRC?

CRC (Cyclic Redundancy Check) is an error-detection technique.

The sender performs **modulo-2 division** using a generator polynomial and appends the remainder to the data.

The receiver performs the same operation to check for errors.

16. What is modulo-2 division in CRC?

Modulo-2 division is binary division in which:

- Addition is performed using **XOR**
- There are no carries or borrows.

17. What is a checksum?

A **checksum** is an error-detection method in which data is divided into fixed-size words, added using one's-complement addition, and the complement of the result is transmitted.

C. HDLC & PPP

18. What is HDLC?

HDLC (High-Level Data Link Control) is a **bit-oriented data link protocol** used for reliable communication.

19. What are the three types of HDLC frames?

HDLC has:

1. **I-frame:** Carries user information.
2. **S-frame:** Used for supervisory/control functions.
3. **U-frame:** Used for unnumbered control functions.

20. What are NRM and ABM in HDLC?

- **NRM (Normal Response Mode):** Secondary station transmits only when permitted by the primary station.
- **ABM (Asynchronous Balanced Mode):** Both stations can communicate independently.

21. What are RR, RNR, REJ and SREJ?

They are HDLC supervisory commands:

- **RR:** Receive Ready
- **RNR:** Receive Not Ready
- **REJ:** Reject
- **SREJ:** Selective Reject

22. What is the P/F bit in HDLC?

The **P/F bit** means **Poll/Final**.

It is used to indicate whether a frame is a poll from the primary station or a final response from the secondary station.

23. What is PPP?

PPP (Point-to-Point Protocol) is a data link protocol used for communication over **point-to-point links**.

It provides framing, error detection and link management.

24. What are LCP and NCP?

- **LCP:** Link Control Protocol; establishes, configures and tests the link.
- **NCP:** Network Control Protocol; configures network-layer protocols.

25. What are PAP and CHAP?

Both provide authentication in PPP.

- **PAP:** Uses a simple username/password exchange.
- **CHAP:** Uses a challenge-response mechanism and is more secure.

D. Multiple Access & ALOHA

26. Why are multiple-access protocols required?

When many devices share the same communication channel, multiple devices may transmit simultaneously.

Multiple-access protocols control channel access and help **reduce collisions**.

27. What are the three categories of multiple-access protocols?

1. **Random access:** ALOHA, CSMA
2. **Controlled access:** Reservation, polling, token passing
3. **Channelization:** FDMA, TDMA, CDMA

28. What is Pure ALOHA?

In **Pure ALOHA**, a station transmits whenever it has data.

If a collision occurs, it waits for a random time and retransmits.

29. What is Slotted ALOHA?

In **Slotted ALOHA**, transmission can begin only at the beginning of a time slot.

This reduces the vulnerable period and improves efficiency compared with Pure ALOHA.

30. What is vulnerable time in ALOHA?

Vulnerable time is the period during which another transmission can cause a collision with a transmitted frame.

For:

- Pure ALOHA = **2T**
- Slotted ALOHA = **T**

where **T is the frame transmission time**.

31. What is CSMA?

CSMA (Carrier Sense Multiple Access) means a station **senses the channel before transmitting**.

If the channel is busy, the station waits according to the persistence method.

32. What are 1-persistent, nonpersistent and p-persistent CSMA?

- **1-persistent:** Transmit immediately when the channel becomes idle.
- **Nonpersistent:** If busy, wait for a random time before sensing again.
- **p-persistent:** Used with slotted channels; transmit with probability **p** when the channel is idle.

E. CSMA/CD & CSMA/CA

33. What is CSMA/CD?

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is used in traditional shared Ethernet.

A station:

1. Senses the channel.
2. Transmits if idle.
3. Detects a collision while transmitting.
4. Stops transmission.
5. Waits using a backoff algorithm.
6. Retransmits.

34. Why is Tfr≥2TpT_{fr} ₩geq 2T_p required in CSMA/CD?

The frame transmission time must be at least twice the **maximum propagation time** so that a sender is still transmitting when a collision from the farthest station can return.

Thus, the sender can **detect the collision**.

35. What is CSMA/CA?

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) is mainly used in wireless networks.

Instead of detecting collisions after they occur, it tries to **avoid collisions before transmission**.

36. Difference between CSMA/CD and CSMA/CA?

CSMA/CD	CSMA/CA
Collision Detection	Collision Avoidance
Used traditionally in Ethernet	Used mainly in WLAN
Detects collision	Tries to avoid collision
Can detect collision while transmitting	Uses mechanisms such as ACK, backoff and RTS/CTS

37. What is the hidden-station problem?

The **hidden-station problem** occurs when two wireless stations cannot hear each other but can both communicate with the same access point.

They may transmit simultaneously and cause a collision.

RTS/CTS can help reduce this problem.

F. Controlled Access

38. What is controlled access?

Controlled access allows stations to access a shared channel in an organized way.

Main methods:

- Reservation

- Polling
- Token passing

39. What is token passing?

A special control frame called a **token** circulates among stations.

Only the station holding the token can transmit, so **collisions are avoided**.

G. MAC Address, ARP & Ethernet

40. What is a MAC address?

A **MAC address** is a physical/link-layer address assigned to a network interface.

A typical Ethernet MAC address is **48 bits**.

MAC address: Used for local/link delivery.
IP address: Used for network-layer delivery.

41. What is ARP?

ARP (Address Resolution Protocol) maps an **IPv4 address to a MAC address** on the local network.

Process:

1. Sender checks its ARP cache.
2. If the MAC is unknown, it broadcasts an ARP request.
3. The device with that IP sends an ARP reply.
4. The sender stores the mapping in its ARP cache.

42. What is Ethernet?

Ethernet is a widely used LAN technology standardized under the **IEEE 802.3** family.

Traditional shared Ethernet used **CSMA/CD**.

43. What are the important fields of an Ethernet frame?

An Ethernet frame contains:

- Preamble
- Start Frame Delimiter
- Destination MAC address
- Source MAC address
- Type/Length
- Data
- Padding, if required
- FCS/CRC

44. Why does Ethernet have a minimum and maximum frame size?

The **minimum frame size** was required for collision detection in traditional shared Ethernet.

The **maximum frame size** limits the amount of data carried in one frame and helps maintain efficient network operation.

45. What is Ethernet efficiency?

Ethernet efficiency indicates how effectively the channel is used for actual data transmission.

It is affected by factors such as:

- Propagation delay
- Frame transmission time
- Network size

A commonly used parameter is:

a=TpTfra=\frac{T_p}{T_{fr}}

where TpT_p is propagation time and TfrT_{fr} is frame transmission time.

46. Difference between hub, switch and router?

Device	Main function
Hub	Sends incoming data to all ports
Switch	Forwards frames using MAC addresses
Router	Forwards packets using IP addresses

47. How does a switch learn MAC addresses?

A switch examines the **source MAC address** of incoming frames and records it in its MAC table along with the incoming port.

If the destination MAC is unknown, the switch generally **floods** the frame to other relevant ports.

48. What are Standard, Fast, Gigabit and 10-Gigabit Ethernet?

They represent different Ethernet speeds:

- **Standard Ethernet:** 10 Mbps
- **Fast Ethernet:** 100 Mbps
- **Gigabit Ethernet:** 1 Gbps
- **10-Gigabit Ethernet:** 10 Gbps

H. Wireless LAN, Bluetooth & WiMAX

49. What is IEEE 802.11? Explain BSS, ESS, DCF and PCF.

IEEE 802.11 is the family of standards for **Wireless LAN (Wi-Fi)**.

- **BSS:** Basic Service Set; a basic wireless network cell.
- **ESS:** Extended Service Set; multiple interconnected BSSs.
- **DCF:** Distributed Coordination Function; contention-based access method.
- **PCF:** Point Coordination Function; centralized access method using a point coordinator.

50. What are Bluetooth and WiMAX?

Bluetooth is a short-range wireless technology used for personal-area networking.

- **Piconet:** Small Bluetooth network formed around a central/master device.
- **Scatternet:** Collection of interconnected piconets.

WiMAX is based on the **IEEE 802.16** family and is designed for broadband wireless communication over larger areas. It can use technologies such as **OFDM/OFDMA and MIMO**.

Module 4 –

Part 1: SNMP and ASN.1

1. What is SNMP?

SNMP stands for Simple Network Management Protocol. It is used to monitor and manage network devices such as routers, switches, and servers.

2. What are the main components of SNMP?

- SNMP Manager: Monitors and manages devices.
- SNMP Agent: Runs on the managed device and responds to requests.
- MIB: Management Information Base, which stores managed-object information.

3. What is the function of an SNMP manager?

It requests information from agents, monitors network performance, and may modify device settings.

4. What is an SNMP agent?

An SNMP agent is software running on a network device that collects management information and responds to the manager.

5. What is MIB?

MIB stands for Management Information Base. It is a collection of managed objects and their definitions used to monitor network devices.

6. What are the main SNMP operations?

- Get: Retrieves a value.
- GetNext: Retrieves the next object in the MIB.
- Set: Changes a value.
- Response: Returns a reply.
- Trap: Sends an alert without waiting for a request.

7. Which transport protocol and port numbers are commonly used by SNMP?

SNMP commonly uses UDP. Port 161 is used for requests and responses, while port 162 is used for traps and notifications.

8. What is ASN.1?

ASN.1 stands for Abstract Syntax Notation One. It is a standard notation used to define the structure and types of data exchanged between systems.

9. What is the relationship between ASN.1 and SNMP?

ASN.1 defines the structure of management data, while encoding rules such as BER specify how that data is represented for transmission.

10. What is BER?

BER stands for Basic Encoding Rules. It converts data defined using ASN.1 into a byte sequence that can be transmitted over a network.

Part 2: Physical Layer – Data and Signals

11. What is the physical layer?

The physical layer is the lowest layer of the OSI model. It transmits raw bits through a physical transmission medium.

12. What is the difference between data and signals?

- Data: The information to be communicated.
- Signal: The electrical, optical, or electromagnetic representation used to carry the data.

13. What are analog and digital signals?

- Analog signal: Changes continuously over time.
- Digital signal: Uses discrete values, commonly represented by 0s and 1s.

14. What are the main characteristics of a periodic signal?

Amplitude, frequency, and phase.

15. What is amplitude?

Amplitude is the maximum strength or magnitude of a signal measured from its reference level.

16. What is frequency?

Frequency is the number of complete cycles of a periodic signal per second. Its unit is hertz (Hz).

17. What is phase?

Phase indicates the relative position of a waveform within its cycle, usually measured in degrees or radians.

18. What is bandwidth?

Bandwidth is the range of frequencies occupied by a signal or supported by a channel. For a frequency range, it is the upper frequency minus the lower frequency.

19. What is the difference between bit rate and baud rate?

- Bit rate: Number of bits transmitted per second.
- Baud rate: Number of signal elements transmitted per second.

For binary signaling, one signal element may carry one bit. With multilevel signaling, one element can carry multiple bits.

20. What is noise? Name its common types.

Noise is unwanted energy that interferes with a signal. Common types include thermal noise, induced noise, crosstalk, and impulse noise.

Part 3: Digital Transmission

21. What is digital transmission?

Digital transmission is the transfer of information using digital signals or digital representations of information.

22. What is line coding?

Line coding converts digital data into a digital signal suitable for transmission over a medium.

23. What is the difference between NRZ-L and NRZ-I?

- NRZ-L: The signal level represents the data bit.
- NRZ-I: A transition at the beginning of a bit interval represents a particular bit value, depending on the convention used.

24. What is Manchester encoding?

Manchester encoding has a transition in the middle of every bit interval. This helps the receiver recover timing information.

25. What is block coding?

Block coding adds redundant bits to groups of data bits to help improve synchronization or detect invalid patterns. For example, 4B/5B maps each 4-bit group to a 5-bit code.

26. What is scrambling?

Scrambling replaces certain bit patterns with predefined patterns to avoid long sequences that may cause synchronization problems.

27. What are B8ZS and HDB3?

They are bipolar scrambling techniques used with AMI signaling to handle long sequences of zeros and maintain synchronization.

28. What is PCM?

PCM stands for Pulse Code Modulation. It converts an analog signal into digital form through sampling, quantization, and encoding.

29. State the sampling theorem.

A band-limited signal with maximum frequency fmf_mfm must be sampled at a rate of at least 2fm2f_m2fm samples per second for ideal reconstruction.

fs≥2fmf_s ₩geq 2f_mfs≥2fm

30. What is quantization error?

Quantization error is the difference between the original sample value and its quantized approximation. For a uniform quantizer, it is typically bounded by half the quantization interval under the usual assumptions.

31. What is delta modulation?

Delta modulation represents the change in signal amplitude between successive samples using a single bit, indicating whether the signal should step up or down.

32. What are slope overload distortion and granular noise?

- Slope overload distortion: The input changes faster than the delta modulator can follow.
- Granular noise: The step size is too large for small variations in the input signal.

Part 4: Analog Transmission

33. What is analog transmission?

Analog transmission sends information using continuously varying analog signals.

34. What is modulation?

Modulation is the process of varying a carrier signal's amplitude, frequency, or phase according to the message signal.

35. Why is modulation required?

Modulation helps place information in a suitable frequency band for transmission, supports channel allocation, and can make practical antenna sizes possible.

36. What is amplitude modulation (AM)?

AM varies the amplitude of the carrier according to the message signal, while the carrier frequency remains constant.

37. What is frequency modulation (FM)?

FM varies the instantaneous frequency of the carrier according to the message signal.

38. What is phase modulation (PM)?

PM varies the phase of the carrier according to the message signal.

39. What is the bandwidth of a conventional AM signal?

For a message signal with maximum frequency BBB, conventional AM has bandwidth:

BAM=2BB_{AM}=2BBAM=2B

It contains the upper and lower sidebands around the carrier.

40. What is the difference between AM, FM, and PM?

Modulation	Carrier property varied
AM	Amplitude
FM	Frequency
PM	Phase

Part 5: Bandwidth Utilization

41. What is bandwidth utilization?

Bandwidth utilization is the efficient use of the available channel bandwidth to transmit information.

42. What is multiplexing?

Multiplexing combines multiple signals or data streams so they can share a common communication link.

43. What is FDM?

FDM stands for Frequency Division Multiplexing. Each signal uses a separate frequency band within the shared channel.

44. What is TDM?

TDM stands for Time Division Multiplexing. Multiple signals share the same channel by transmitting in different time slots.

45. What is WDM?

WDM stands for Wavelength Division Multiplexing. It allows multiple optical signals at different wavelengths to travel through the same optical fiber.

46. What is spread spectrum?

Spread spectrum distributes a signal over a wider frequency band than the minimum required. It can improve resistance to interference and support multiple-access systems.

Part 6: Transmission Media

47. What are transmission media?

Transmission media are the physical paths or wireless environments through which signals travel from sender to receiver.

48. What are guided and unguided media?

- Guided media: Signals travel through a physical path, such as twisted-pair cable, coaxial cable, or optical fiber.
- Unguided media: Signals travel through free space using electromagnetic waves, such as radio or microwave signals.

49. What is the difference between twisted-pair cable, coaxial cable, and optical fiber?

Medium	Main characteristic
Twisted pair	Low cost; commonly used in telephone and Ethernet networks
Coaxial cable	Shielded construction; offers better protection from interference than basic unshielded twisted pair
Optical fiber	Carries light; supports high bandwidth and is immune to electromagnetic interference

50. What are the differences between radio waves, microwaves, and infrared waves?

- Radio waves: Often used for broadcasting and wireless communication; can provide broad coverage.
- Microwaves: Commonly used in point-to-point links, satellite communication, and cellular networks.
- Infrared waves: Used for short-range communication and generally do not penetrate walls effectively.

Quick revision: formulas to remember

Topic	Formula
Bandwidth	B=fmax − fmin B=f_{₩max}-f_{₩min}B=fmax− fmin
Sampling theorem	fs≥2fmf_s₩geq 2f_mfs≥2fm
Bits per sample	n=log2Ln=\log_2 Ln=log2L
PCM bit rate	Rb=fs×nR_b=f_s\times nRb=fs×n
AM bandwidth	BAM=2BB_{AM}=2BBAM=2B
Signal-to-noise ratio in dB	SNRdB=10log10(S/N)SNR_{dB}=10\log_{10}(S/N)SNRdB=10log10(S/N)

Module 1

Part 1: Overview of the Internet

1. What is the Internet?

The Internet is a global network of interconnected computer networks that communicate using the TCP/IP protocol suite.

2. What is a computer network?

A computer network is a group of interconnected devices that communicate and share data and resources.

3. What is an end system or host?

An end system is a device connected to the Internet that sends or receives data, such as a computer, smartphone, or server.

4. What is an ISP?

ISP stands for Internet Service Provider. It provides users and organizations with access to the Internet.

5. What is a protocol?

A protocol is a set of rules that defines how devices communicate, including message format, order, and actions taken when messages are sent or received.

6. What are the basic components of a network?

The main components are end systems, communication links, packet switches such as routers, and protocols.

7. What is packet switching?

Packet switching divides data into smaller units called packets. These packets are forwarded through the network toward their destination.

8. What is the difference between a router and a switch?

A router forwards packets between IP networks, while a network switch usually forwards Ethernet frames within a local area network using MAC addresses.

Part 2: Protocol Layering

9. What is protocol layering?

Protocol layering divides network communication into separate layers, with each layer performing specific functions.

10. Why is layering used in computer networks?

Layering simplifies network design, supports modular development, and allows one layer to change without redesigning the entire system, provided its interfaces remain compatible.

11. Name the five layers of the Internet protocol stack.

1. Application layer
2. Transport layer
3. Network layer
4. Link layer
5. Physical layer

12. What is the function of the application layer?

It provides network services to applications, such as web browsing, file transfer, e-mail, and name resolution.

13. What is the function of the transport layer?

It provides communication between application processes on different hosts. TCP and UDP are important transport-layer protocols.

14. What is the function of the network layer?

It delivers packets from the source host to the destination host across interconnected networks. IP is its principal protocol.

15. What are encapsulation and decapsulation?

- Encapsulation: Each layer adds its control information to the data as it moves down the protocol stack.
- Decapsulation: The receiving system removes and processes this information as data moves up the stack.

16. What is the difference between TCP/IP and the OSI model?

The TCP/IP model describes the practical protocol architecture used on the Internet. The OSI model is a seven-layer reference model used to understand network functions.

Part 3: Application-Layer Paradigms

17. What is the application layer?

The application layer is the top layer of the Internet protocol stack. It enables network applications to exchange messages.

18. What is an application-layer protocol?

It defines the format, meaning, and exchange rules for messages between application processes. Examples include HTTP, FTP, SMTP, and DNS.

19. What are the two main application architectures?

The two main architectures are the client-server paradigm and the peer-to-peer (P2P) paradigm.

20. What is the client-server paradigm?

A server provides services or resources, while clients request and use those services.

21. What is the peer-to-peer paradigm?

In P2P communication, devices called peers can communicate directly and share resources without depending entirely on a dedicated central server.

22. What is the difference between client-server and P2P architectures?

Client-server	P2P
Uses dedicated servers	Peers share resources
Server provides services to clients	Each peer may act as both client and server
Server may become a bottleneck	Workload can be distributed among peers
Example: traditional web service	Example: BitTorrent

Part 4: World Wide Web and HTTP

23. What is the World Wide Web (WWW)?

The WWW is a collection of interlinked web resources that users access over the Internet using web browsers and web protocols.

24. What is a web browser?

A web browser is client software that requests, receives, and displays web content. Examples include Chrome, Firefox, and Edge.

25. What is a web server?

A web server stores or generates web resources and sends them to clients in response to requests.

26. What is HTTP?

HTTP stands for Hypertext Transfer Protocol. It is an application-layer protocol used to transfer web resources between clients and servers.

27. What is the difference between HTTP and HTTPS?

HTTP transfers web data without built-in transport encryption. HTTPS uses HTTP over TLS to provide encryption, integrity protection, and server authentication.

28. What are HTTP request and response messages?

- Request: Sent by a client to ask for a resource or perform an operation.
- Response: Sent by the server with a status code, headers, and sometimes a message body.

29. What are GET and POST methods in HTTP?

- GET: Requests a representation of a resource.
- POST: Submits data for processing, such as form data.

30. Is HTTP a stateful or stateless protocol?

HTTP is fundamentally stateless: each request is understood independently. Applications can maintain user sessions using cookies and other mechanisms.

Part 5: FTP and Electronic Mail

31. What is FTP?

FTP stands for File Transfer Protocol. It is an application-layer protocol used to transfer files between a client and a server.

32. Which transport protocol does FTP use?

FTP uses TCP, which provides reliable, connection-oriented data transfer.

33. What are the control and data connections in FTP?

FTP traditionally uses a control connection for commands and replies and a separate data connection for transferring files or directory listings.

34. What is electronic mail?

Electronic mail, or e-mail, is a system for sending and receiving digital messages over a network.

35. What are the main components of an e-mail system?

- User agent: used to compose and read e-mails.
- Mail server: stores and forwards e-mails.
- Protocols: transfer messages and allow users to retrieve them.

36. What are SMTP, POP3, and IMAP?

- SMTP: Used to send and relay e-mail.
- POP3: Used to retrieve e-mail, commonly by downloading messages.
- IMAP: Used to access and manage messages stored on a mail server.

37. What is the difference between SMTP and HTTP?

SMTP is primarily used to send and relay e-mail messages, whereas HTTP is used to transfer web resources and support web applications.

Part 6: DNS

38. What is DNS?

DNS stands for Domain Name System. It translates domain names, such as `example.com`, into associated information such as IP addresses.

39. Why is DNS required?

DNS lets people use memorable domain names instead of having to remember numerical IP addresses.

40. What is a DNS resolver?

A DNS resolver obtains DNS information on behalf of a client, either from its cache or by querying other DNS servers.

41. What are the main levels in the DNS hierarchy?

The main levels include root DNS servers, top-level domain (TLD) servers, and authoritative DNS servers.

42. What is the difference between recursive and iterative DNS queries?

- Recursive query: The contacted DNS server is asked to obtain the final answer or return an error.
- Iterative query: A server may return a referral to another DNS server that can help resolve the name.

43. Which transport protocol and port are commonly used by DNS?

DNS commonly uses UDP port 53 for ordinary queries and responses. TCP port 53 is also used, including for zone transfers and responses that require TCP.

Part 7: P2P Networks and BitTorrent

44. What is a P2P network?

A peer-to-peer network allows participating devices to share resources directly with one another.

45. What are the advantages and disadvantages of P2P networks?

Advantages: Resource sharing, scalability in many settings, and reduced dependence on a single server.

Disadvantages: Peers may disconnect, availability can vary, and security and coordination can be challenging.

46. What is BitTorrent?

BitTorrent is a P2P file-distribution protocol that lets users download different parts of a file from multiple peers.

47. What is a torrent file or torrent metadata?

It contains information that helps a BitTorrent client identify the content and locate or coordinate with peers. Modern BitTorrent systems may use magnet links instead of a separate `.torrent` file.

48. What is a chunk in BitTorrent?

A chunk is a piece of a file distributed among peers. A client can download different chunks from different peers and assemble them into the complete file.

49. What is a tracker in BitTorrent?

A tracker helps peers discover other peers participating in a particular torrent. It generally coordinates peer discovery rather than storing the entire file.

50. What is the difference between a seed and a leecher in BitTorrent?

- Seed: A peer that has the complete content and uploads pieces to other peers.
- Leecher: A peer that is still downloading the content; it may also upload pieces it already has.

Quick revision: Important full forms

Abbreviation	Full form
ISP	Internet Service Provider
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
HTTP	Hypertext Transfer Protocol
HTTPS	HTTP Secure
FTP	File Transfer Protocol
SMTP	Simple Mail Transfer Protocol
POP3	Post Office Protocol version 3
IMAP	Internet Message Access Protocol
DNS	Domain Name System
P2P	Peer-to-Peer
WWW	World Wide Web

Module 2

Part 1: Transport Layer – Services and Protocols

1. What is the transport layer?

The transport layer provides logical communication between application processes running on different hosts.

2. What are the main services provided by the transport layer?

Process-to-process delivery, multiplexing, demultiplexing, and—depending on the protocol—reliable data transfer, flow control, and congestion control.

3. What is the difference between the transport layer and network layer?

The network layer provides host-to-host delivery, whereas the transport layer provides process-to-process delivery.

4. What are transport-layer protocols?

They are protocols that govern data transfer between application processes. The two main Internet transport protocols are TCP and UDP.

5. What is multiplexing and demultiplexing?

- Multiplexing: Collecting data from multiple application sockets for transmission.
- Demultiplexing: Delivering received segments to the appropriate application socket.

6. What is a socket?

A socket is a software interface through which an application sends and receives data over a network.

7. What is a port number?

A port number identifies an application endpoint and helps the transport layer deliver data to the correct process. It is a 16-bit number.

Part 2: UDP – User Datagram Protocol

8. What is UDP?

UDP stands for User Datagram Protocol. It is a connectionless transport protocol that sends datagrams without guaranteeing delivery or ordering.

9. What are the main characteristics of UDP?

- Connectionless communication
- Small header
- No built-in retransmission
- No built-in flow control
- No built-in congestion control

10. What is the size of the UDP header?

The UDP header is 8 bytes long.

11. What are the fields in a UDP header?

Source port, destination port, length, and checksum. Each field occupies 16 bits.

12. Why is UDP called connectionless?

UDP does not establish a connection or perform a handshake before sending datagrams.

13. What is the purpose of the UDP checksum?

It helps detect errors in a UDP segment. It does not correct errors or retransmit lost data.

14. Give examples of applications that use UDP.

DNS queries, real-time audio/video applications, and many online games use UDP.

Part 3: Reliable Data Transfer and TCP

15. What is reliable data transfer?

Reliable data transfer ensures that data reaches the receiving application correctly and in order, using mechanisms to handle problems such as loss and corruption.

16. What is an acknowledgment (ACK)?

An ACK is a message that confirms the receipt of data or indicates the next data expected by the receiver.

17. What is a timeout in reliable data transfer?

A timeout occurs when an expected acknowledgment does not arrive within a specified time. The sender may retransmit the data.

18. What is TCP?

TCP stands for Transmission Control Protocol. It provides connection-oriented, reliable, ordered delivery of a byte stream.

19. What is the TCP three-way handshake?

TCP establishes a connection using three segments:

1. SYN — the client requests a connection.
2. SYN-ACK — the server acknowledges and responds.
3. ACK — the client acknowledges the response.

20. What is the purpose of sequence numbers in TCP?

Sequence numbers identify positions of bytes in the TCP byte stream and help detect missing data and maintain ordering.

21. What is the purpose of acknowledgment numbers in TCP?

The acknowledgment number indicates the sequence number of the next byte expected by the receiver.

22. What is the TCP sliding-window mechanism?

It allows multiple bytes to be transmitted before acknowledgments arrive, improving efficiency while limiting outstanding data.

23. What is TCP flow control?

TCP flow control prevents the sender from overwhelming the receiver. It uses the receiver-advertised window to limit unacknowledged data.

24. What is the difference between TCP and UDP?

TCP	UDP
Connection-oriented	Connectionless
Reliable and ordered byte stream	Best-effort datagram delivery

Provides flow and congestion control	No built-in flow or congestion control
Minimum header: 20 bytes	Header: 8 bytes
Used for many web and file-transfer connections	Used by DNS and many real-time applications

Part 4: Congestion Control

25. What is network congestion?

Congestion occurs when traffic exceeds available network resources, causing increased delay, queue buildup, and packet loss.

26. What is congestion control?

Congestion control adjusts sending rates and manages traffic to reduce network overload.

27. What is the difference between flow control and congestion control?

- Flow control: Protects the receiver from receiving more data than it can handle.
- Congestion control: Helps prevent excessive traffic from overloading the network.

28. What is the TCP congestion window (cwnd)?

The congestion window is a sender-side limit on the amount of unacknowledged data TCP can have in the network.

29. What is slow start in TCP?

Slow start is a congestion-control phase in which TCP increases its congestion window rapidly, typically exponentially per round-trip time under ideal acknowledgment behavior.

30. What is congestion avoidance in TCP?

Congestion avoidance is a phase in which TCP generally increases its congestion window more gradually, often approximately linearly per round-trip time.

Part 5: Network Layer – Introduction and Forwarding

31. What is the network layer?

The network layer delivers packets from a source host to a destination host across interconnected networks.

32. What are the main functions of the network layer?

Logical addressing, routing, forwarding, and delivering packets across interconnected networks.

33. What is the difference between forwarding and routing?

- Forwarding: Moving a packet from an input interface to the appropriate output interface.
- Routing: Determining the path packets should take through the network.

34. What is a forwarding table?

A forwarding table contains information that helps a router choose the outgoing interface or next hop for a packet.

35. What is store-and-forward transmission?

A router receives the complete packet before transmitting it onto the next link. This introduces transmission delay at each store-and-forward hop.

36. What is IP?

IP stands for Internet Protocol. It provides logical addressing and best-effort packet delivery across interconnected networks.

Part 6: IP Addressing and Next-Generation IP

37. What is an IP address?

An IP address is a logical address assigned to a network interface to enable communication across IP networks.

38. What is the difference between IPv4 and IPv6?

IPv4	IPv6
32-bit address	128-bit address

Dotted-decimal notation	Hexadecimal colon-separated notation
Example: `192.168.1.1`	Example: `2001:db8::1`
Variable-length header	Fixed 40-byte base header

39. What is CIDR?

CIDR stands for Classless Inter-Domain Routing. It represents network prefixes using slash notation, such as `192.168.1.0/24`.

40. What is a subnet mask?

A subnet mask identifies the network-prefix bits and host bits in an IPv4 address.

41. What is the purpose of a default gateway?

A default gateway forwards packets to destinations outside the local network when no more specific route applies.

Part 7: Unicast, Multicast, and Routing

42. What is unicast routing?

Unicast routing delivers packets from one source to one destination.

43. What is multicast routing?

Multicast routing delivers packets from a source to a selected group of receivers rather than requiring separate source transmissions for every receiver.

44. What is the difference between unicast, broadcast, and multicast?

- Unicast: One sender to one receiver.
- Broadcast: One sender to all devices in the relevant broadcast domain, where supported.
- Multicast: One sender to a selected group of receivers.

45. What is IGMP?

IGMP stands for Internet Group Management Protocol. In IPv4, it allows hosts and directly connected multicast routers to manage multicast group membership.

46. What is intra-domain routing?

Intra-domain routing finds routes within a single autonomous system. Examples include RIP and OSPF.

47. What is inter-domain routing?

Inter-domain routing exchanges reachability information between different autonomous systems. BGP is the main Internet inter-domain routing protocol.

48. What is the difference between distance-vector and link-state routing?

Distance-vector	Link-state
Exchanges route information with neighboring routers	Distributes link-state information within the routing area
Uses neighbor information to calculate routes	Builds a topology map and calculates shortest paths
Example: RIP	Example: OSPF

Part 8: Quality of Service (QoS)

49. What is Quality of Service (QoS)?

QoS refers to techniques used to manage network performance to meet the requirements of different traffic types.

50. What are the main QoS parameters?

- Bandwidth: Available data-carrying capacity.
- Delay: Time taken for data to reach its destination.
- Jitter: Variation in packet delay.
- Packet loss: Proportion of packets that fail to reach the destination.

Important full forms for quick revision

Abbreviation	Full form
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
IP	Internet Protocol
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
ACK	Acknowledgment
SYN	Synchronize
CIDR	Classless Inter-Domain Routing
IGMP	Internet Group Management Protocol
RIP	Routing Information Protocol
OSPF	Open Shortest Path First

BGP	Border Gateway Protocol
QoS	Quality of Service

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